claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_49aab888-e10\agent-a02694afcee8d791a.jsonl`
- SHA-256: `90109cea14e07c7646e22bfb02c9bb42fd34e4e977431a795ae5a8f805e4bb2f`
- Source modified: `2026-06-08T13:08:14+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_49aab888-e10`
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`

Transcript

1. user (2026-06-08T13:04:39.587Z)

DIMENSION = COST + LICENSING + PRIVACY/COMPLIANCE. Use WebSearch + WebFetch. (a) Concrete training COST estimates: fine-tuning a small TAL head (cheap) vs fine-tuning a 2-7B video VLM (GPU-hours x price) ? give $ ranges per option using current cloud GPU prices. (b) LICENSING: confirm which candidate BASE models are truly Apache-2.0/MIT (and which are NOT, e.g. Gemma 3, Qwen-72B-tier, Llama community license, any NC dataset); and that training on our OWN footage + human labels is clean. (c) PRIVACY/COMPLIANCE for a product that processes user match video: data-residency (India), GDPR (no minors, biometrics future), and which platforms contractually DON'T train on customer data. Cite URLs for license + privacy-term claims.

Keep it aligned to OUR situation:
OUR SITUATION (a local-first racket-sports highlight indexer + rally detector):
- GOAL: an OWNED, on-device-capable model for RALLY TEMPORAL SEGMENTATION (predict rally [start,end] intervals in a match video), evolving toward "Option B" = an OWNED END-TO-END MULTIMODAL model (frames + audio + pose -> rallies/highlights) fine-tuned on our own data. Privacy lever = local inference, user video never leaves the device.
- HARD GUARDRAILS (non-negotiable): (1) base/teacher models for TRAINING must be OPEN Apache-2.0 (or MIT/BSD) ONLY; video-capable bases like Qwen-VL / Gemma family are candidates BUT **Gemma 3 is REJECTED (viral/non-standard license)**. (2) NEVER use paid LLM outputs (Gemini/OpenAI/Anthropic) as TRAINING data ? they are inference/eval/teacher-for-eval only (terms + copyright). (3) Exclude minors from the training pool; biometrics/gait = future (GDPR Art. 9). (4) Generic data schema; coach->athlete is an optional overlay.
- DATA ASSETS WE WANT TO REUSE FOR TRAINING: human-labeled rally [start,end] CSVs per match (currently Adarsh-and-Avi 96 rallies, Kushagra 32, more coming; schema rally_number,start_time,end_time,ending_reason,sport) + per-rally ending_reason labels (winner/forced_error/unforced_error/service_fault/let) + per-video WASB shuttle-trajectory CSVs (Frame,Visibility,X,Y; from a frozen MIT-licensed shuttle detector) + (eval-only, NOT training) Gemini silver labels. Footage = single-camera amateur GoPro, 1080p/4K, 60fps.
- CURRENT PIPELINE: WASB (frozen vision shuttle detector) -> trajectory -> hand-tuned windowing + a net-crossing "gate" + a small distilled LogisticRegression on trajectory features. Honest results: threshold tuning ceilings ~F1 0.73 vs HUMAN labels on one match; a learned model UNDERPERFORMS baseline at n=2 videos. Bottleneck = more golden data + WASB recall, not the method.
- CONSTRAINTS: solo founder, BANGALORE-based (cost-sensitive; cloud GPU pricing + data-residency matter), local hardware = ONE RTX 2070 Super (8GB) + WSL2. User explicitly mentioned Vertex AI but wants MORE options + a detailed comparison.
- THE TASK CLASS is temporal action localization/segmentation (action = "rally"); labels are segment intervals.

Grounding brief:
{
  "owned_model_goal": "Build an OWNED, on-device-capable generative video-QA model (fine-tuned from an Apache-2.0 video VLM base) for RALLY TEMPORAL SEGMENTATION and future end-to-end multimodal sport understanding. Privacy lever: local inference, user video never leaves device. The model plugs into the existing provider registry with paid Gemini as an always-ready

fallback. Long-term path (Option B): collapse modular local cues (WASB trajectory, audio hit detection, pose detection) into a single fine-tuned end-to-end model. Near-term (Option A, already shipping): modular signal fusion on frozen or near-frozen bases. The moat is NOT the model weights ? it's the consented badminton-specific video?QA dataset + the eval harness. Owned model becomes primary only after scaffolding (async jobs, multi-tenant, storage/compute abstractions) is rock-solid and the golden-set eval/training corpus is healthy.",
  "constraints": [
    "Base model MUST be Apache-2.0 clean (commercial-viable, fine-tunable, derivative-wrappable). REJECT Gemma 3 (viral Google PUP), Llama Vision (700M-MAU cap + naming mandate). APPROVED bases: Qwen2.5-VL-7B/32B (Apache-2.0 at those sizes only; 3B/72B custom-licensed), Qwen3-VL-4B/8B, Gemma 4 multimodal (Apache-2.0, 2026), InternVL2.5 (MIT).",
    "HARD RULE: Training data must be Apache-2.0 or MIT open-model outputs ONLY. FORBIDDEN: any outputs from paid frontier APIs (Gemini, OpenAI, Anthropic, Claude) as training signal ? they are competing-model clauses and breach terms. Paid APIs are inference/evaluation/teacher-reference only, never training data sources. This is non-negotiable per legal.",
    "Exclude minors' video from the training pool BY DEFAULT. Training requires separate, explicit opt-in consent (distinct from processing consent). Per-user training adapters trained only on that user's own consented video; deletion = drop the adapter.",
    "Generic data schema (no coach-athlete hardcoding). Coach/athlete overlay is optional, orthogonal. Data model stays reusable for solo users, academies, power users.",
    "Training evaluation must use a held-out golden QA eval set (controlled, curated, not user-contributed). Agent-runnable training requires trustworthy pass/fail verdict (not eyeballing). Regression testing locked to golden labels to prevent metric inflation from data leakage.",
    "Video-QA fine-tuning works ONLY on labeled data. Currently available: 96 (Adarsh+Avi) + 32 (Kushagra) human-labeled rally [start,end] CSVs per match + Gemini silver labels (eval-only, not training). 5k?20k curated clip?QA pairs is the realistic training band."
  ],
  "data_assets": [
    "Human-labeled rally [start,end] CSVs per match (Adarsh-and-Avi 96 rallies, Kushagra 32; schema: rally_number, start_time, end_time, ending_reason, sport). Source: sports-data-collector repo, curate export with commercial-license gate.",
    "Per-rally ending_reason labels (winner, forced_error, unforced_error, service_fault, let) ? rally semantics feeding the QA-pair dataset.",
    "Per-video WASB shuttle-trajectory CSVs (Frame, Visibility, X, Y) from frozen MIT-licensed WASB-SBDT shuttle detector. Trajectory is a primary local signal for multi-cue fusion.",
    "Gemini silver labels (eval and teacher-reference ONLY, explicitly not training-data). Used to measure oracle quality and as a training-time prior (Option A modular fusion), never harvested as outputs to train the owned model.",
    "Single-camera amateur GoPro footage: 1080p/4K, 60fps. 22/22 ShuttleSet matches re-acquired at best available resolution (Phase 2 complete). ~26 GB corpus licensed/owned, no user uploads.",
    "Golden-set ground-truth expansion via Roora vendor (Bengaluru, initial pilot underway): 3-sport pilot (badminton, pickleball, table tennis) with validated rally CSV schema and annotation guideline.",
    "Sibling rallyannotator repo (MIT, open-source, VLC-based): owner can self-produce expert-labeled eval clips. Feed via `import-local` to the collector and `--labels` loaders to the indexer."
  ],
  "task_framing": "**Task: design a training study blueprint for the OWNED video-QA model that clarifies (1) what data is safe/available now, (2) which base model to lock, (3) which fine-tuning backend fits our constraints, and (4) how to stage the transition from heuristics ? Option A modular fusion ? Option B end-to-end owned model.**\\n\\n**Current pipeline (heuristics + frozen detectors):** WASB (frozen vision) ? trajectory ? hand-tuned windowing + net-crossing gate + distilled LogisticRegression on trajectory features. Honest baseline: threshold-tuned F1 ? 0.73 vs human labels (one match); learned model underperforms at n=2 videos. Bottleneck = golden data + WASB recall, not method.\\n\\n**Option A (now ? 6?12 months):** Modular, fully-local cues: WASB trajectory + audio hit-detection (E1 probe AMBER / leaning negative, gated on more golden labels) + pose/court serve-gating (validated but heavy). Each cue swappable, independently testable, generates training signal for the end-to-end model. Fusion happens in our rally layer, not inside frozen detectors.\\n\\n**Option B (12?24+ months, after scaffolding + healthy corpus):** Collapse modular cues into a single owned multimodal rally model fine-tuned on (frames + audio + pose/stance + ?) ? rally [start, end]. Removes recall ceiling imposed by WASB candidate windows and licensing question on academic-trained weights. Requires: (a) enough golden eval/training video (owner self-labels via rally-annotator + vendor labels), (b) end-to-end training harness (nanochat-style operability: one command, report card, deterministic,

reproducible), (c) a measured improvement bar vs. the modular Option A.\\n\\n**Bridge data flow (cross-repo, ADR-002 contract):** sports-data-collector's per-match rally CSVs + manifest.json (commercial-license gate) ? indexer's golden eval set and training signal. Collector's 5-sport tables (badminton/tennis/cricket/pickleball/table-tennis) share a (video_id, ordinal, start_time, end_time) shape, so the bridge is sport-generic. Golden-set expansion funded via Roora (Bengaluru) ? initial 3-sport pilot.\\n\\n**Immediate blockers on training-study readiness:** (1) **Owned-model base selection + licensing lock** (Qwen vs Gemma 4; verify per-checkpoint commercial terms). (2) **Fine-tuning backend decision** (self-managed ms-swift on rented GPU [RunPod/Lambda/Modal] vs. managed Together/Fireworks vs. enterprise Predibase). (3) **Golden QA eval harness design** (the crux: nanochat-style pass/fail verdict). (4) **Training data schema & consent** (how to convert rally CSVs + ending-reason labels + Gemini reference ? clip?QA pairs without using paid outputs; minors exclusion). (5) **Data leakage guardrails** (`train | eval` split by match, not clip; overlap detection). (6) **Training job orchestration** (how it slots into the async job model P0, runs on ComputeBackend, fits BYO-compute).\\n\\n**Success criteria for the study:** (a) Blueprint for converting existing golden labels into 5k?20k curated clip?QA pairs. (b) Confirmed Apache-2.0 base + verified commercial DPA from the fine-tune backend. (c) Proof-of-concept golden QA eval gate that blocks a regression-quality model. (d) Staged roadmap: Option A (modular, 2?4 weeks) ? Option B e2e model (after corpus health signals, 3?6 months post-scaffold). (e) Clear cost/latency comparison (self-hosted 7B VLM vs. Gemini per-call on target workload).\"",
  "roadmap_alignment_notes": [
    "**Sequencing principle (owner directive):** Make scaffolding (P0?P4 in PLATFORM_ARCHITECTURE §6) rock-solid FIRST, keep the known paid stack (Gemini) as an always-ready fallback, and only then shift focus to owned-model quality (P5). This study unblocks P5's design but does NOT start implementation until P0?P4 infrastructure is delivered.",
    "**P0 (async job model, single-tenant)** is the prerequisite: owned-model fine-tuning runs as an orchestration job on ComputeBackend GPUs (§5A / DEFERRED #16). Without async + job dispatch, training is not operationalizable.",
    "**P1?P3 (multi-tenant, storage/compute abstraction)** enable the data flywheel: training corpus lives in Storage (§3.1), training/inference jobs dispatch to ComputeBackend (§3.2), per-user adapters live in the compute tier. Without these, the owned model is stranded on a single box.",
    "**P4 (rich rally/shot semantic metadata)** feeds P5: the annotation registry + candidate_segments + golden-set labels are the training-signal source. Per-rally ending_reason labels + human-correction loop ? QA-pair generation.",
    "**C14 (COMMERCIALIZATION.md)** tracks the owned-model licensing + training-data policy as a P1 blocker: Apache-2.0 base lock, open-model teachers only, no paid-LLM training signal, separate explicit training opt-in, minors excluded. This study must resolve C14 before any code lands.",
    "**ADR-004 substrate decision** (2026-06-02 update): Phase 4 offline segmenter is WASB shuttle-motion, not YOLO player-motion (YOLO @ IoU-0.5 recall too low). The owned end-to-end model (Option B) will eventually subsume WASB's role. ADR-007 audio fusion is the Option A stepping stone.",
    "**ADR-007 direction (accepted):** Audio hit-detection is the next rally-detection cue (Option A). It is fully local, permissive, and fuses cleanly with the distilled classifier. Gated on golden labels (E1 probe AMBER, awaits more data). This is the realistic 2?4 week Option A milestone while waiting for scaffold + corpus growth.",
    "**Golden-set expansion (ADR-006 / Roora pilot):** Vendor-produced labels are the highest-leverage move. With Roora on 3-sport badminton/pickleball/table-tennis, the training corpus grows month-on-month. The training study should plan for incremental re-training (CI-integrated, agent-runnable) as each batch lands.",
    "**Long-term moat (§5A, PLATFORM):** The defensible assets are (a) the consented, badminton-specific video?QA dataset and (b) the eval harness. Do NOT message 'we own an LLM' ? the base VLM (Qwen/Gemma) is freely downloadable. Message the dataset + eval precision instead.",
    "**Cost & margin (C7 / COMMERCIALIZATION):** Self-hosted 7B VLM (?16 GB VRAM) on rented GPU is cheaper per inference than Gemini only if: (a) sustained QA volume is high, (b) training corpus is defensibly large, (c) per-call API cost pain is real. Below those thresholds, paid API is cheaper than standing GPU + second stack. Benchmark before committing budget.",
    "**Training harness operability (§5A, PLATFORM):** The goal is 'an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled video and get a trustworthy ship/no-ship verdict, unattended.' This requires: (1) one command, end-to-end (ms-swift engine, recipe-driven); (2) one complexity dial (auto-derived defaults); (3) automatic report card (held-out golden QA eval vs target); (4) reproducible + deterministic + cheap; (5) minimal/readable. Nanochat is the reference for operability (not the engine). The harness = thin glue over ms-swift + the golden eval + the provider registry."

  ]
}

2. assistant (2026-06-08T13:04:41.959Z)

I'll research this dimension thoroughly. Let me start by loading the web tools and checking relevant project docs.

3. user (2026-06-08T13:04:43.036Z)

docs\COMMERCIALIZATION.md

4. assistant (2026-06-08T13:04:45.790Z)

Let me read the key project docs and start parallel web research.

5. user (2026-06-08T13:04:46.771Z)

```
1       # Commercialization Tracker
2
3       **Status:** ? LIVING ? track this in parallel with product/quality work.
4       **Owner:** avidullu · **Last refreshed:** 2026-06-07
5
6       **Purpose.** Single source of truth for the **commercial** dimension of the project:
7       licensing/legal posture, codec & output economics, hosting unit-economics, the
8       golden-set vendoring spend, packaging/distribution, and the product-market-fit
9       risks that gate monetization. Updated as decisions land ? see the
10      [Changelog](#changelog) at the bottom.
11
12      **Merged from / supersedes:**
13      - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14        audit) ? kept as the **research-of-record** (per-claim verification transcript,
15        primary sources). This doc carries its *current* status forward.
16      - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17        archived; its commercial findings + the quality-phase priorities live here now.
18
19      **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20      users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21      distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22      **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23      royalties** all still apply.
24
25      ---
26
27      ## 1. Commercialization dashboard
28
29      Legend: ? done · ? in progress / watch · ? blocker / needs action · ?? needs legal
counsel · ? future
30      Priority: **P0** (do first) · **P1** · **P2** (lowest). Rows are grouped **Pending ? In-
flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
order.)
31
32      | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
33      |---|---|---|---|---|---|
34      | | **? PENDING (needs action / not started)** | | | | |
35      | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
36      | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37      | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
```

`PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md` (#16). |
	? IN-FLIGHT (in progress / watch)				
C4	**Runtime AI strategy (A/B/C + owned model)**	**P0**	?	Live = **(C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry.	Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A.
C13	**Beachhead validation**	**P0**	?	Market research (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated.	Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See §6.
C8	**Golden-set vendoring spend**	**P1**	?	ADR-006: **Roora (Bengaluru)** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors).	Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN_SET_VENDORING.md`.
C3	**Shuttle-tracking weights provenance**	**P1**	? ??	WASB-SBDT **code is MIT (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code grant. MonoTrack/YOLO-NAS banned (noncommercial).	Train-own weights (Phase-4 roadmap) **or** confirm per-checkpoint terms ?? before commercial deploy.
C14	**Owned-model base licensing + training-data policy**	**P1**	? ??	Owned video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ? **Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented dataset + eval harness**, not the model.	Lock to Apache-2.0; **open-model teachers only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7.
C6	**Inbound HEVC decode**	**P2**	? ??	Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty.	Low/uncertain residual; confirm decode-side exposure with counsel before launch.
C7	**Hosting unit-economics**	**P2**	?	Per-video cost is *cheap* for Gemini (~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured.	**Benchmark $/video & latency on the target VM** for A/B/C before pricing. See §5.
C10	**Value layer (product surface)**	**P2**	?	Minimal exports shipped (EDL/JSON/CSV + `basic_stats`, PR #51).	Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6.
C11	**Input-quality enforcement**	**P2**	?	Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50).	Turn more `VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules.
C2	**`supervision` ByteTrack pin**	**P2**	?	Pinned `<0.30.0` (deprecation ahead).	Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one.
	? COMPLETED				
C1	**AGPL / copyleft removal**	?	?	ADR-005 shipped: AGPL `ultralytics` YOLOv8 replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL code/weights in the stack.	Done. Keep new deps permissive.

---

2. Licensing & dependency posture (refreshed risk table)

Verdicts updated against the current stack (`requirements.txt` + code). Original
2026-05-30 verdicts and the 25-claim verification transcript are in
`archives/MONETIZATION_AUDIT.md`.

Dependency	License	SaaS obligation?	Current verdict	Notes
~~`ultralytics` + `yolov8n.pt`~~	AGPL-3.0	Yes (§13)	? **REMOVED**	Replaced per ADR-005 ? no longer in the stack.
torchvision detectors (Faster R-CNN/RetinaNet)	BSD-3	No	? LIVE	The chosen replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`).
`supervision` (ByteTrack)	MIT	No	? LIVE	Pinned `<0.30.0` ? migrate before

bump (C2). |
WASB-SBDT (shuttle)	MIT (code)	No	? code clear	Weights provenance open (C3).
MonoTrack / YOLO-NAS	noncommercial	?	? BANNED	Do not use.
Gemini API (paid tier)	service terms	Paid: no training/review	? LIVE WATCH	Must bill the paid tier to keep user video private (C4).
H.264 / HEVC / AAC codecs	patent pools	royalties on encode/decode	? ??	Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6).
ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv	LGPL/Apache/BSD/MIT	No	? CLEAR	Invoked as subprocess / imported; all permissive.
`obsreport` (reports)	permissive (private)	No	? soft dep	Pin the `git+ssh@vX` tag once the repo is published (BACKLOG).

Bottom line: the two hard blockers from the audit (Ultralytics; banned weights) are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)** and **weights provenance (C3)**, plus the metered-API privacy posture (C4).

---

3. Runtime AI strategy ? A/B/C

The semantic rally-boundary step is the one external/metered dependency. Three paths (from the audit), with current standing:

- **(A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
- **(B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
- **(C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native; ongoing metered cost + external dependency. **? current live path.**

Hosted alternatives for (C) comparison: Anthropic Claude (no training on inputs/outputs; customer owns outputs; **no native video** ? weaker here); OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.

---

4. Codec / output economics (the live gap ? C5)

The trap: ffmpeg being free (LGPL) does **not** grant the *patent* rights to the codecs it implements. Royalties attach to the act of encode/decode.

- **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA) is first 100k units/yr royalty-free with a permanent "free to end-users over the internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
- **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing must use **SVT-AV1 (CPU)**.
- **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools, highest uncertainty; low residual.

Recommendation: provision an **Ada-class GPU (L4/L40)** and standardize highlight output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely. *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*

---

5. Hosting unit-economics (C7)

Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.

- **(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263

tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
- **(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
- **(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training + light GPU. **Best steady-state margin.**

> The A/B/C $/video comparison depends on GPU instance + batch throughput ?
> **benchmark on the target VM before pricing the product.**

---

6. Product-market fit & the quality/data phase

Beachhead (from market research, 2026-06-07 ? see [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
The static-camera constraint is a **B2B filter**: the buyer must already record from a fixed angle, which points away from the median casual player and toward coaches/clubs.
Ranked beachheads:
1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+ players, formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches who already monetize, badminton whitespace vs incumbents. Sell to the **academy**, not the individual; needs INR-tuned pricing.
2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8% YoY) but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ? head-to-head, not greenfield.
3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China needs local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.

Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/Spiideo; comparable to SwingVision software-only).

PMF risks (from the readiness review)
- **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld footage is a failure mode. The wedge (static, tripod, full-court) is defensible but narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
- **Value beyond raw segments is table stakes.** Time-saved editing alone is weak; per-rally tags, stats, and a correction loop are what make it sticky (C10).
- **Golden labels are the recurring blocker** across the offline segmenter, audio fusion, and measurable quality ? the highest-leverage investment (C8).

Refined priority order (quality/data phase ? "focus on quality once new videos land")
Carried from the readiness review; commercialization items above are tracked in parallel.

1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos): ingest the new batch, expand calibration/CV/regression baselines, make label contribution + safe retraining easy and non-contaminating (eval/training separation).
2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics land.
3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/ min-duration + `rally_gate`; validate WASB on real video; trial audio fusion (ADR-007).
4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ? `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ? validators/report rules; "your video is good enough" UX.
6. **Multi-sport maturation + recording best practices.**

```
7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
8. **Polish & distribution** (C9): installer story, first-run path, codec/output, economics.

**Recommendation:** with the mid-week labeled-video batch, spend ~80% on #1 + #3
(flywheel + quality iteration on real data), then unification (#2) and deeper value (#4).

---

## 7. Open questions needing owner / counsel sign-off ??

1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves
   insufficient (fallback). Quote-only.
2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
   confirm each (C3).
3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
   figures with counsel before relying on the "small SaaS under threshold" reading (C5/C6).
4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
   throughput (C7).
5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec switch.)

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## Changelog

- **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
  ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
  "beachhead validation" workstream. Wired the summary into §6.
- **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
  audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
  removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
  the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
  `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
  exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.

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## See also
- `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors, beachhead; cited).
- `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources + verification transcript).
- `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
- `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor process).
- `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
- `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
- `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
- `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.
```

6. user (2026-06-08T13:04:53.803Z)

Web search results for query: "RunPod Lambda Labs cloud GPU pricing per hour H100 A100 RTX 4090 2026"

Links: [{"title":"GPU Cloud Pricing Comparison 2026: Lambda, RunPod, Vast.ai, AWS Live Rates","url":"https://altstreet.investments/tools/gpu/gpu-price-comparison"},{"title":"GPU Cloud Pricing | Per-Second H100, A100, RTX | Runpod","url":"https://www.runpod.io/pricing"},{"title":"GPU Cloud Pricing 2026: H100 from $1.03/hr, B200 from $2.12/hr (15+ providers) | Spheron Blog","url":"https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/"},{"title":"Cloud GPU Pricing 2026: A100 $1.99/hr, H100 $3.29/hr+ | SynpixCloud","url":"https://www.synpixcloud.com/blog/cloud-gpu-pricing-comparison-2026"},{"title":"RunPod vs Lambda Labs vs CoreWeave: GPU Cloud Pricing","url":"https://www.buildmvpfast.com/blog/gpu-cloud-cost-comparison-runpod-lambda-labs-coreweave-2026"},{"title":"RunPod Review 2026 - Cheapest H100 GPU Pricing and Serverless Guide | DeployBase","url":"https://deploybase.ai/articles/runpod-review"},{"title":"AI Cloud Pricing | GPU Compute & AI Infrastructure | Lambda","url":"https://lambda.ai/pricing"},{"title":"RunPod GPU Pricing 2026 ? H100 & A100 Costs","url":"https://gpucost.org/provider/runpod"},{"title":"Runpod GPU pricing: A complete breakdown and platform comparison | Blog ? Northflank","url":"https://northflank.com/blog/runpod-gpu-pricing"}]

Based on the search results, here's a comprehensive breakdown of cloud GPU pricing for RunPod and Lambda Labs in 2026:

RunPod Pricing (as of May 2026)

H100 PCIe starts at $1.99/hr on-demand, H100 SXM runs $2.69/hr, and A100 80GB ranges from $1.19-1.39/hr depending on configuration. RTX 4090s cost just $0.34.

RunPod spot market offers 40-50% discounts for interruptible compute, with a spot H100 costing $1.30-$1.60/hr.

Lambda Labs Pricing (as of April-May 2026)

Lambda has 1x H100 PCIe at $3.29/hr and 1x H100 SXM at $4.29/hr. A100 80GB SXM in 8-GPU systems is $2.79/GPU/hr, while A100 PCIe instances are listed at $1.99/hr.

Key Comparison

RunPod still has the cheapest self-serve H100s in a three-way comparison at $1.99/hr for H100 PCIe and $2.69/hr for H100 SXM, with Lambda Labs pricing significantly higher at the same tier.

Note that RunPod adds per-instance fees covering compute resources and storage, typically $0.10-$0.40 depending on configuration, so total costs will be higher than GPU rates alone.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-06-08T13:04:56.165Z)

I have good context. Let me run the core research searches in parallel for cost, licensing, and privacy.

8. user (2026-06-08T13:05:02.284Z)

Web search results for query: "Qwen2.5-VL 7B Apache 2.0 license Hugging Face fine-tuning commercial"

Links: [{"title":"wangkanai/qwen2.5-vl-7b-instruct · Hugging Face","url":"https://huggingface.co/wangkanai/qwen2.5-vl-7b-instruct"},{"title":"Qwen2.5 VL 7B Instruct ? Try & Compare | Roboflow Playground","url":"https://playground.roboflow.com/models/qwen/qwen2-5-vl-7b-instruct"},{"title":"Qwen/Qwen2.5-VL-7B-Instruct · Qwen2.5-VL-7B-Instruct License?","url":"https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/discussions/31"},{"title":"Qwen/Qwen2.5-VL-7B-Instruct · Hugging Face","url":"https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct"},{"title":"Qwen/Qwen2.5-72B-Instruct · Question: Clarification on Qwen License Agreement Regarding Output Usage","url":"https://huggingface.co/Qwen/Qwen2.5-72B-Instruct/discussions/18"},{"title":"Qwen2.5: A Party of Foundation Models! | Qwen","url":"https://qwenlm.github.io/blog/qwen2.5/"},{"title":"LICENSE · Qwen/Qwen2.5-7B at main","url":"https://huggingface.co/Qwen/Qwen2.5-7B/blob/main/LICENSE"},{"title":"Qwen/Qwen2.5-VL-7B-Instruct at main","url":"https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/tree/main"},{"title":"aws-prototyping/OmniLong-Qwen2.5-VL-7B · Hugging Face","url":"https://huggingface.co/aws-prototyping/OmniLong-Qwen2.5-VL-7B"},{"title":"Qwen/Qwen2.5-7B-Instruct · Hugging Face","url":"https://huggingface.co/Qwen/Qwen2.5-7B-Instruct"}]

Based on the web search results, here's what I found about Qwen2.5-VL 7B licensing and commercial use:

License Information

Qwen2.5-VL-7B-Instruct is a 7-billion parameter vision-language model from Alibaba's QwenLM team, released on January 26, 2025 under the Apache 2.0 license. The Apache 2.0 license allows both commercial and non-commercial use with attribution.

Fine-tuning and Commercial Use

The model is available on Hugging Face and is suitable for fine-tuning. Weights are publicly available on Hugging Face and GitHub, making it suitable for both research and applied multimodal use.

The Apache 2.0 license permits commercial applications of fine-tuned models, though users should note that while the base model remains governed by Apache 2.0, a different license may be applied to the modifications introduced during the fine-tuning phase, provided that this does not infringe upon the terms set forth by the original license.

This makes Qwen2.5-VL-7B a commercially-friendly option for organizations looking to fine-tune and deploy vision-language models under permissive licensing terms.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-06-08T13:05:02.867Z)

Web search results for query: "Together AI Fireworks fine-tuning pricing per token DPA no training on customer data 2026"

Links: [{"title":"A complete guide to Together AI pricing in 2026 | eesel AI","url":"https://www.eesel.ai/blog/together-ai-pricing"},{"title":"Fireworks - Pricing","url":"https://fireworks.ai/pricing"},{"title":"Together.ai API Pricing 2026 ? All Models, Live Rates | AI Pricing Guru","url":"https://www.aipricing.guru/together-pricing/"},{"title":"Fireworks AI Alternatives 2026: Best Replacements for Serverless LLM Inference and Fine-Tuning | Spheron Blog","url":"https://www.spheron.network/blog/fireworks-ai-alternatives/"},{"title":"Fireworks AI vs Together AI Pricing 2026 ? Model & Cost Comparison | Price Per Token","url":"https://pricepertoken.com/endpoints/compare/fireworks-vs-together"},{"title":"Fireworks AI vs Together AI (2026): Features, Pricing & ...","url":"https://www.respan.ai/market-map/compare/fireworks-ai-vs-together-ai"},{"title":"Pricing | Together AI","url":"https://www.together.ai/pricing"},{"title":"Fireworks AI Pricing 2026: Free-$9/per million tokens / hour","url":"https://costbench.com/software/llm-api-providers/fireworks-ai/"},{"title":"Together AI Pricing In 2026: Models, Costs, And How To Manage Your Bill","url":"https://www.cloudzero.com/blog/together-ai-pricing/"},{"title":"Together AI vs Fireworks: Pricing, Speed and Benchmarks 2026 | DeployBase","url":"https://deploybase.ai/articles/together-ai-vs-fireworks-pricing-speed-and-benchmark"}]

Based on my search results, here's what I found about Together AI and Fireworks fine-tuning pricing in 2026:

Fine-Tuning Pricing Per Token

Together AI: LoRA SFT on a 16B base is $0.48/M tokens; full fine-tuning on a 70-100B base is $3.20/M; DPO on a 70-100B is $8.00/M.

Fireworks AI: Fine-tuning starts at USD 0.50 per 1M tokens for models up to 16B parameters.

DPA (Data Processing Agreement) and Training on Customer Data

I could not find specific information in the search results about Data Processing Agreements (DPA) or explicit policies stating "no training on customer data" for either Together AI or Fireworks AI as of 2026. The search results focused primarily on pricing structures and technical capabilities rather than data privacy commitments or DPA terms.

For information about DPA terms and data handling policies, you would likely need to:
- Contact the providers' sales or legal teams directly
- Review their official terms of service and privacy policies
- Check their enterprise agreements

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. assistant (2026-06-08T13:05:05.387Z)

Let me verify the Qwen license details directly from Hugging Face, and check the Gemma 4 / InternVL licensing plus Vertex AI training data terms.

11. user (2026-06-08T13:05:09.800Z)

License Information for Qwen/Qwen2.5-VL-7B-Instruct

Based on the webpage content, the model uses the **Apache 2.0 license**, as indicated by the label "License: apache-2.0" displayed on the model card.

The provided documentation does not include explicit statements about commercial use restrictions or fine-tuning permissions beyond what is inherent to the Apache 2.0 license framework. For detailed terms governing commercial applications or model modifications, you would need to consult the full license text or the model's official documentation on the Qwen GitHub repository.

12. user (2026-06-08T13:05:12.846Z)

Web search results for query: "Qwen3-VL 4B 8B 30B license Apache 2.0 which sizes custom license 2026"

Links: [{"title":"qwen3:8b/license","url":"https://ollama.com/library/qwen3:8b/blobs/d18a5cc71b84"},{"title":"Qwen3: Think Deeper, Act Faster | Qwen","url":"https://qwenlm.github.io/blog/qwen3/"},{"title":"Qwen 3: The new open standard - by Nathan Lambert","url":"https://www.interconnects.ai/p/qwen-3-the-new-open-standard"},{"title":"Qwen3-VL-8B-Instruct VRAM Requirement + Unsloth Fine-Tuning (Patch Size, trust_remote_code)","url":"https://kaitchup.substack.com/p/qwen3-vl-fine-tuning-on-your-computer"},{"title":"Qwen Models Guide: 600M to 1 Trillion Parameters","url":"https://www.digitalapplied.com/blog/qwen-models-complete-guide"},{"title":"GitHub - QwenLM/Qwen3-VL: Qwen3-VL is the multimodal large language model series developed by Qwen team, Alibaba Cloud. · GitHub","url":"https://github.com/QwenLM/Qwen3-VL"},{"title":"BeDiscovER: The Benchmark of Discourse Understanding in the Era of Reasoning Language Models","url":"https://arxiv.org/pdf/2511.13095"},{"title":"What Users Leave Unsaid: Under-Specified Queries Limit Vision-Language Models","url":"https://arxiv.org/pdf/2601.06165"},{"title":"LICENSE · Qwen/Qwen3-8B at main","url":"https://huggingface.co/Qwen/Qwen3-8B/blob/main/LICENSE"},{"title":"Qwen3-VL/LICENSE at main · QwenLM/Qwen3-VL","url":"https://github.com/QwenLM/Qwen3-VL/blob/main/LICENSE"}]

Based on the search results, here's what I found about Qwen3-VL model sizes and licensing:

Qwen3-VL Model Sizes and Releases

The Qwen3-VL-4B (Instruct/Thinking) and Qwen3-VL-8B (Instruct/Thinking) were released on October 15, 2025. The Qwen3-VL-30B-A3B-Instruct and Qwen3-VL-30B-A3B-Thinking were released on October 4, 2025. Additionally, the Qwen3-VL-235B-A22B-Instruct and Qwen3-VL-235B-A22B-Thinking were released on September 23, 2025.

Licensing

The Qwen3-VL models are available on the Hugging Face Hub with an Apache 2.0 license. The search results indicate that all the available Qwen3-VL models (4B, 8B, 30B, and 235B sizes) use the Apache 2.0 license, with no mention of custom licenses for specific sizes.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. assistant (2026-06-08T13:05:15.769Z)

Let me verify the Qwen2.5-VL-3B and 72B licensing specifically (the carve-out claim), and Vertex AI / Gemini training data terms, plus Modal pricing.

14. user (2026-06-08T13:05:18.537Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://cloud.google.com/vertex-ai/generative-ai/docs/data-governance
Redirect URL: https://docs.cloud.google.com/vertex-ai/generative-ai/docs/data-governance
Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://docs.cloud.google.com/vertex-ai/generative-ai/docs/data-governance"
- prompt: "Does Google use customer prompts, inputs, or outputs from Vertex AI / Gemini API to train or improve its models? What is the default data governance policy for paid Vertex AI generative AI? Quote any statement about not using customer data for training."

15. user (2026-06-08T13:05:20.632Z)

Web search results for query: "Qwen2.5-VL-3B-Instruct license qwen-research non-commercial 72B custom license"

Links: [{"title":"LICENSE · Qwen/Qwen2.5-VL-3B-Instruct at main","url":"https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE"},{"title":"LICENSE · Qwen/Qwen2.5-VL-3B-Instruct at acbce9a6c80e81356680a632166380c8cc1d27ec","url":"https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/acbce9a6c80e81356680a632166380c8cc1d27ec/LICENSE"},{"title":"Qwen2.5 VL 3B | Open Laboratory","url":"https://openlaboratory.ai/models/qwen2_5-vl-3b-instruct"},{"title":"LICENSE · Qwen/Qwen2.5-3B at main","url":"https://huggingface.co/Qwen/Qwen2.5-3B/blob/main/LICENSE"},{"title":"LICENSE · Qwen/Qwen2.5-VL-3B-Instruct at 9c1a731d62346b46f7e75419f9f1f1a0dbf1decb","url":"https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/9c1a731d62346b46f7e75419f9f1f1a0dbf1decb/LICENSE"},{"title":"LICENSE · Qwen/Qwen2.5-3B-Instruct at main","url":"https://huggingface.co/Qwen/Qwen2.5-3B-Instruct/blob/main/LICENSE"},{"title":"Qwen2.5 VL 3B Instruct - API Pricing & Providers | OpenRouter","url":"https://openrouter.ai/qwen/qwen2.5-vl-3b-instruct"},{"title":"LICENSE · Qwen/Qwen2.5-VL-72B-Instruct at main","url":"https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE"},{"title":"LICENSE · Qwen/Qwen2.5-32B-Instruct at main","url":"https://huggingface.co/Qwen/Qwen2.5-32B-Instruct/blob/main/LICENSE"}]

Based on the search results, Qwen2.5-VL-3B-Instruct weights are released under a Qwen Research License, and cannot be utilized for commercial purposes. You should read the license to verify if your use case is permitted.

The search results also indicate that the Qwen2.5-VL series includes larger 7B and 72B parameter variants, with the 3B model being designed for efficient deployment. However, the specific license details for the 72B variant and information about custom licensing were not fully detailed in the search results.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

16. assistant (2026-06-08T13:05:22.977Z)

Confirmed the 3B carve-out. Let me fetch the Vertex AI redirect, verify the 72B license, and check Qwen2.5-VL-72B license text.

17. user (2026-06-08T13:05:29.866Z)

Qwen License Analysis

This is the **Qwen custom license**, not Apache 2.0.

Commercial Use Restrictions

Yes, there is a significant monthly active user threshold. Section 4 states:

> "If you are commercially using the Materials, and your product or service has more than 100 million monthly active users, you shall request a license from us."

In other words, commercial users exceeding 100 million monthly active users must obtain explicit authorization from Alibaba Cloud to continue use.

Key Points

- **License type**: Proprietary Qwen License Agreement (effective September 19, 2024)
- **Basic rights**: Non-exclusive, worldwide, royalty-free license for use, reproduction, and derivative works
- **Attribution requirement**: Users must display "Built with Qwen" or "Improved using Qwen" in product documentation
- **Restriction trigger**: Commercial products/services with 100+ million monthly active users need additional permission

18. user (2026-06-08T13:05:30.276Z)

Google's Data Usage Policy for Vertex AI

Does Google use customer data to train models?

No. Google has a clear commitment against this practice:

> "Google won't use your data to train or fine-tune any AI/ML models without your prior permission or instruction."

This policy applies to all managed models on Vertex AI, including both general availability and pre-GA models.

Default Data Governance Policy

The default policy ensures minimal data retention. Google caches customer inputs and outputs in-memory only (not stored at rest) with a 24-hour expiration to improve performance. This caching:
- Remains isolated at the project level
- Adheres to data residency requirements
- Does not violate zero data retention principles
- Can be disabled if needed

Limited Exceptions

Vertex AI maintains some data for specific purposes (abuse monitoring, grounding with Google Search/Maps, and Gemini Live session resumption), but customers can request exceptions or use alternative services to achieve complete zero data retention if required.

19. assistant (2026-06-08T13:05:33.761Z)

I've confirmed the Qwen size-by-size license carve-outs precisely. Now let me check the remaining base models (Gemma 4, InternVL2.5, Gemma 3 confirmation) and the fine-tuning backend DPA terms (Together/Fireworks/Modal/Predibase), plus India data residency.

20. user (2026-06-08T13:05:39.394Z)

Web search results for query: "InternVL2.5 license MIT commercial use Hugging Face OpenGVLab"

Links: [{"title":"OpenGVLab/InternVL2_5-Pretrain-Models · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2_5-Pretrain-Models"},{"title":"About the license of internVL2.5-4B model · Issue #900 · OpenGVLab/InternVL","url":"https://github.com/OpenGVLab/InternVL/issues/900"},{"title":"OpenGVLab/InternVL2-Pretrain-Models · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2-Pretrain-Models"},{"title":"OpenGVLab/InternVL2_5-8B-MPO · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2_5-8B-MPO"},{"title":"OpenGVLab/InternVL2_5-2B · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2_5-2B"},{"title":"OpenGVLab/InternVL2_5-4B-MPO · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2_5-4B-MPO"},{"title":"OpenGVLab/InternVL3_5-2B · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL3_5-2B"},{"title":"OpenGVLab/InternVL2_5-4B-AWQ · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2_5-4B-AWQ"},{"title":"OpenGVLab/InternVL2_5-1B-MPO · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2_5-1B-MPO"},{"title":"OpenGVLab/InternVL2-1B · Hugging Face","url":"https://huggingface.co/OpenGVLab/InternVL2-1B"}]

Based on the search results, here's what I found about InternVL2.5 licensing:

License Information

InternVL2.5 is released under the MIT license, which is a permissive open-source license that allows commercial use.

Important Consideration About Commercial Use

However, there's an important caveat to note regarding commercial use. Some InternVL2.5 models use pre-trained components like Qwen2.5-3B-Instruct, which is licensed under the Apache License 2.0.

There was a specific concern raised in the GitHub community: some InternVL2.5 models components may use Qwen versions with a RESEARCH LICENSE that prohibits commercial use.

Recommendation

While the InternVL2.5 project itself is MIT-licensed (allowing commercial use), the specific model variant you plan to use may incorporate pre-trained components with different licensing terms. It's important to verify the license of the particular InternVL2.5 variant you intend to use on Hugging Face or the OpenGVLab GitHub repository to ensure all its components are compatible with your intended commercial use case.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. user (2026-06-08T13:05:43.059Z)

Web search results for query: "Gemma terms of use "prohibited use policy" distribution restrictions viral license commercial 2026"

Links: [{"title":"Gemma Terms of Use | Google AI for

Developers","url":"https://ai.google.dev/gemma/terms"},{"title":"google/gemma-7b · How are you handling Gemma?s license for commercial apps?","url":"https://huggingface.co/google/gemma-7b/discussions/122"},{"title":"'Open' AI model licenses often carry concerning restrictions | TechCrunch","url":"https://techcrunch.com/2025/03/14/open-ai-model-licenses-often-carry-concerning-restrictions/"},{"title":"Google Gemma: The Hidden Risks of an ?Almost Open? License - WCR.LEGAL","url":"https://wcr.legal/google-gemma-license-risks/"},{"title":"Gemma Prohibited Use Policy | Google AI for Developers","url":"https://ai.google.dev/gemma/prohibited_use_policy"},{"title":"Gemma License (danger) is not Free Software and is not Open Source","url":"https://gnu.support/gnu-emacs/emacs-lisp/Gemma-License-danger-is-not-Free-Software-and-is-not-Open-Source.html"},{"title":"What Is Gemma 4's Apache 2.0 License? Why It Matters More Than the Model Itself | MindStudio","url":"https://www.mindstudio.ai/blog/gemma-4-apache-2-license-commercial-use"},{"title":"Gemma License ? CanIRun.ai","url":"https://www.canirun.ai/license/gemma"},{"title":"What Is the Gemma 4 Apache 2.0 License? Why It Changes Everything for Commercial AI Deployment | MindStudio","url":"https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment"}]

Based on the search results, here's what I found regarding Gemma's terms of use, prohibited use policy, distribution restrictions, and licensing:

Gemma Terms of Use Overview

Gemma's terms incorporate a Prohibited Use Policy by reference at ai.google.dev/gemma/prohibited_use_policy.

Distribution Restrictions

To reproduce or distribute Gemma or Model Derivatives, users must include use restrictions as an enforceable provision in any agreement governing use/distribution, provide notice to recipients, and furnish third party recipients with a copy of the Gemma Terms of Use. Additionally, all non-hosted distributions must be accompanied by a notice stating "Gemma is provided under and subject to the Gemma Terms of Use found at ai.google.dev/gemma/terms".

Prohibited Use Policy

The Prohibited Use Policy restricts several categories of use, including generating and distributing content intended to misinform or mislead, promoting spam, and generating content for deceptive or fraudulent activities. It also prohibits generating content that infringes, misappropriates, or violates any individual's or entity's rights, including copyrights.

Commercial Use

The Terms explicitly permit commercial use?you can build products on Gemma, charge users, deploy in SaaS products or enterprise software, and there is no revenue threshold triggering a separate commercial arrangement (unlike Llama 3's 700M MAU clause).

Important Note: Gemma 4

Gemma 4, released in 2025, comes under Apache 2.0, with the license covering the model weights themselves. Earlier Gemma versions used the custom "Gemma Terms of Use" referenced above.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

22. assistant (2026-06-08T13:05:46.832Z)

This confirms the Gemma 3 (custom viral terms) vs Gemma 4 (Apache-2.0) distinction. Let me now nail down the fine-tuning backend DPA/training terms and Together AI India residency, plus verify the Gemma 4 Apache-2.0 claim more carefully (it's a key 2026 claim).

23. user (2026-06-08T13:05:52.523Z)

Together AI Fine-Tuning Pricing

Standard Pricing (Per 1M Tokens)

Supervised Fine-Tuning (LoRA):
- Up to 16B: $0.48
- 17B-69B: $1.50
- 70-100B: $2.90

Full Fine-Tuning:
- Up to 16B: $1.20
- 17B-69B: $3.75
- 70-100B: $7.25

Direct Preference Optimization (DPO) - LoRA:
- Up to 16B: $0.54
- 17B-69B: $1.65
- 70-100B: $3.20

DPO - Full Fine-Tuning:
- Up to 16B: $1.35
- 17B-69B: $4.12
- 70-100B: $8.00

Specialized Model Pricing

The platform offers specialized pricing for proprietary models (DeepSeek, GLM, Kimi, Qwen, Llama variants) ranging from $3?$100 per million tokens depending on the model and method.

Data Privacy

The provided content contains no statements regarding data privacy or assurances about not training on customer data.

24. user (2026-06-08T13:06:02.220Z)

Web search results for query: "Together AI Fireworks privacy policy "do not train" customer data fine-tuning HIPAA SOC2 zero retention"

Links: [{"title":"Fireworks.ai Achieves SOC 2 Type II and HIPAA Compliance","url":"https://fireworks.ai/blog/fireworks-ai-achieves-soc-2-type-ii-and-hipaa-compliance"},{"title":"Data Security - Fireworks AI Docs","url":"https://docs.fireworks.ai/guides/security_compliance/data_security"},{"title":"Powering Secure AI: Together AI Achieves SOC 2 Type 2 Compliance","url":"https://www.together.ai/blog/soc-2-compliance"},{"title":"Fireworks.ai Achieves SOC 2 Type II and HIPAA Compliance","url":"https://www.linkedin.com/posts/fireworks-ai_fireworksai-achieves-soc-2-type-ii-and-hipaa-activity-7123691351606538241-LcJo"},{"title":"SOC 2 for AI Companies: Complete Guide (2025) | Comp AI","url":"https://www.trycomp.ai/hub/soc-2-for-ai-companies"},{"title":"From Zero to Certified: A Practical Guide to SOC 2, HIPAA, ...","url":"https://www.linkedin.com/pulse/from-zero-certified-practical-guide-soc-2-hipaa-gdpr-compliance-tuck-gbrqc"},{"title":"AI-Powered SOC 2 Evidence Collection Explained | Censinet","url":"https://censinet.com/perspectives/ai-powered-soc-2-evidence-collection-explained"},{"title":"Is Fireworks AI SOC 2 Compliant? 2026 Security Status | RiscLens","url":"https://risclens.com/compliance/directory/fireworks-ai"},{"title":"Fireworks AI: Trust Center","url":"https://trust.fireworks.ai/"},{"title":"7 Best SOC 2 Compliant AI Support Platforms for Regulated Industries [2026 Guide] | Fini Labs","url":"https://www.usefini.com/guides/best-soc-2-compliant-ai-support-platforms-regulated-industries-2026"}]

Let me search for more specific information about the "do not train" policy and data retention practices.

Links: [{"title":"Is My Customer Data Used to Train the AI Model? | Twig |
Twig","url":"https://www.twig.so/blog/is-customer-data-used-to-train-ai-
model"},{"title":"Training, Fine-Tuning, and Inference | Systems | A Thinking
Workspace","url":"https://pruningmypothos.com/systems/training-vs-inference/"},{"title":"Fine-
Tuning Privacy: What Happens to Your Custom Training
Data","url":"https://stealthcloud.ai/intelligence/fine-tuning-privacy-custom-
training/"},{"title":"Fine-Tuned Models Remember Everything: The Training Data Privacy Problem -
DEV Community","url":"https://dev.to/tiamatenity/fine-tuned-models-remember-everything-the-
training-data-privacy-problem-4a9e"},{"title":"Navigating the Designs of Privacy-Preserving
Fine-tuning for Large   Language
Models","url":"https://arxiv.org/pdf/2501.04323"},{"title":"Inference-Aware Fine-Tuning for
Best-of-N Sampling in Large Language
Models","url":"https://arxiv.org/pdf/2412.15287"},{"title":"AI Inference vs Training vs Fine-
Tuning","url":"https://shieldbase.ai/blog/ai-inference-vs-training-vs-fine-tuning"}]

Now let me search specifically for Together AI and Fireworks privacy policies regarding data retention and training.

Links: [{"title":"Privacy Policy | Together
AI","url":"https://www.together.ai/privacy"},{"title":"Terms of Service | Together
AI","url":"https://www.together.ai/terms-of-service"},{"title":"Zero Data
Retention","url":"https://vercel.com/docs/ai-gateway/capabilities/zdr"},{"title":"Most Private
AI for Lawyers: Why Zero Data Retention Wins in 2026 -
Spellbook","url":"https://www.spellbook.legal/learn/most-private-ai"},{"title":"Privacy-first AI
in Europe: zero retention, sovereignty, and ...","url":"https://regolo.ai/privacy-first-ai-in-
europe-zero-retention-sovereignty-and-the-new-risks-we-cannot-ignore/"},{"title":"Zero data
retention in AI: what it really means and how enterprises can enforce it with Regolo -
regolo.ai","url":"https://regolo.ai/zero-data-retention-vs-provider-policies-what-teams-need-to-
verify/"},{"title":"Zero Data Retention Enforcement for AI Agents: The New Standard for
Enterprise Trust | NeuralTrust","url":"https://neuraltrust.ai/blog/zero-data-retention-
agents"},{"title":"Privacy Policy | xAI","url":"https://x.ai/legal/privacy-
policy"},{"title":"Cursor · Data Use & Privacy Overview","url":"https://cursor.com/data-use"}]

Links: [{"title":"Fireworks - Privacy Policy","url":"https://fireworks.ai/privacy-
policy"},{"title":"Zero Data Retention - Fireworks AI
Docs","url":"https://docs.fireworks.ai/guides/security_compliance/data_handling"},{"title":"Data
Security - Fireworks AI Docs","url":"https://fireworks.ai/docs/guides/security_compliance/data_s
ecurity"},{"title":"Fireworks AI: Trust
Center","url":"https://trust.fireworks.ai/"},{"title":"Fireworks.ai Achieves SOC 2 Type II and
HIPAA Compliance","url":"https://fireworks.ai/blog/fireworks-ai-achieves-soc-2-type-ii-and-
hipaa-compliance"},{"title":"Trust Center - Fireworks
AI","url":"https://trust.fireworks.ai/faq"},{"title":"Fireworks AI Data Privacy Concerns -
Integration Diagnosis","url":"https://drdroid.io/integration-diagnosis-knowledge/fireworks-ai-
data-privacy-concerns"},{"title":"AI Agent Policy | Terms of Service and Usage
Guidelines","url":"https://www.elitefireworks.com/pages/ai-agent-terms"},{"title":"Fireworks -
Terms of Service","url":"https://fireworks.ai/terms-of-service"},{"title":"GitLab Duo data usage
| GitLab Docs","url":"https://docs.gitlab.com/user/gitlab_duo/data_usage/"}]

Search Results Summary

Based on my web search, here's what I found regarding Together AI and Fireworks regarding privacy policies, data retention, and compliance:

Together AI

Together AI does not use any data collected from you to train models without explicit opt-in and consent, and allows you to tweak privacy settings in the platform so that they do not retain training data, prompts, or model responses for any purpose.

Under Zero Data Retention (ZDR), a setting that can be enabled by users, data and outputs are not stored, retained, or used for model training, product improvements, or any other purpose beyond the time required to process the specific request or transaction.

Together AI adheres to HIPAA requirements, including data encryption in transit and at rest, audit logging, and strict business associate agreements (BAAs) with partners.

Fireworks AI

Fireworks AI does not use customer Content - including prompts, inputs, outputs, or other request/response Content submitted through services - to train, fine-tune, or improve any Fireworks foundation models.

Fireworks has Zero Data Retention by default, specifically meaning Fireworks does not log or store prompt or generation data for any open models, without explicit user opt-in. Prompt and generation data exist only in volatile memory for the duration of the request, and are not logged into any persistent storage.

If a customer uses Fireworks AI's fine-tuning or training services, any datasets uploaded for that purpose are used only to train the customer's specified model and remain under the customer's control.

Fireworks.ai inference platform is both SOC 2 Type II and HIPAA compliant, and Fireworks is HIPAA compliant and supports healthcare and life sciences organizations in leveraging rapid inference capabilities with confidence.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

25. assistant (2026-06-08T13:06:05.928Z)

Strong privacy findings on both backends. Let me verify the Gemma 4 Apache-2.0 claim from a primary source (it's a 2026 release and load-bearing), and check India data residency / DPDP Act for the privacy angle.

26. user (2026-06-08T13:06:12.063Z)

Gemma 4 Apache 2.0 License Status

License Type:
Yes, according to the article, Gemma 4 uses Apache 2.0. The piece states: "Gemma 4 ships under a true Apache 2.0 license?no custom restrictions, no compete clauses."

Release Date:
The article indicates Gemma 4 was "released in April 2025," though the blog post itself is dated April 9, 2026.

Prohibited Use Policy:
The article does not mention any current Prohibited Use Policy attached to Gemma 4. It explicitly contrasts Gemma 4's Apache 2.0 approach with earlier versions that "included content restrictions," but indicates these were replaced entirely by the standard Apache 2.0 license.

Multimodal/Video Capabilities:
Gemma 4 is multimodal for images. The article describes "the 27B multimodal variant" as "accepting both text and image inputs" and notes it "handles image understanding tasks." However, there is no mention of video capabilities in the provided content.

Note on Specificity:
The article does not reproduce the actual Apache 2.0 license text or direct quotes from Google's official Gemma 4 documentation. Its claims about the licensing are framed as editorial assertions rather than verbatim license language.

27. user (2026-06-08T13:06:18.201Z)

Web search results for query: "India DPDP Act 2023 data residency video personal data GDPR minors consent processing cloud"

Links: [{"title":"FAQs - The Digital Personal Data Protection Act, 2023","url":"https://www.cyrilshroff.com/wp-content/uploads/2025/12/FAQs-DPDPA.pdf"},{"title":"India?s Digital Personal Data Protection Act, 2023: A Transformative Step in Data Privacy Law - Lexology","url":"https://www.lexology.com/library/detail.aspx?g=e5a2f362-20b1-4e69-ba5e-5ee9e5eb195c"},{"title":"THE DIGITAL PERSONAL DATA PROTECTION ACT, 2023 (NO. 22 OF 2023)","url":"https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf"},{"title":"DPDP Act 2023: India's Digital Personal Data Protection Law Explained","url":"https://www.matters.ai/compliance/dpdp/dpdp-act-2023"},{"title":"India Data Privacy Laws: DPDP Act 2023 and DPDP Rules 2025 Complete Guide | Recording Law","url":"https://www.recordinglaw.com/world-laws/world-data-privacy-laws/india-data-privacy-laws/"},{"title":"Everything You Need to Know About the DPDP Act India, 2023 - Knovos","url":"https://www.knovos.com/blog/everything-you-need-to-know-about-the-dpdp-act-india/"},{"title":"The Digital Personal Data Protection Act, 2023 Advent of Privacy Era in India","url":"https://www.ey.com/content/dam/ey-unified-site/ey-com/en-in/insights/cybersecurity/documents/ey-india-dpdp-act-2023.pdf"},{"title":"125+ DPDPA FAQ | Digital Personal Data Protection Act 2023 & DPDP Rules 2025","url":"https://www.dpdpa.com/dpdpa-faq.html"},{"title":"The Digital Personal Data Protection Bill, 2023","url":"https://prsindia.org/billtrack/digital-personal-data-protection-bill-2023"},{"title":"India Digital Personal Data Protection Act (DPDPA 2025): Updated Guide - CookieYes","url":"https://www.cookieyes.com/blog/india-digital-personal-data-protection-act-dpdpa/"}]

Based on the web search results, here's comprehensive information about the India DPDP Act 2023 covering your query topics:

Overview of the DPDP Act 2023

India's Digital Personal Data Protection Act, 2023 (DPDP Act) marks a transformative shift in India's data governance, replacing the limited IT Act, 2000 and SPDI Rules, 2011 with a comprehensive, consent-driven framework that applies to all digital personal data, including processing by government and entities abroad targeting Indian residents.

Data Residency & Cross-Border Transfers

Unlike GDPR's strict approach, the Act takes a deliberately permissive approach to cross-border data transfer, allowing personal data to be transferred outside India to any country, except those that the Central Government explicitly restricts by notification (a 'negative list' approach). As of February 2026, no countries have been added to this restricted list. However, for Significant Data Fiduciaries, the government retains the power to mandate localisation of specific categories of personal data and traffic data, based on recommendations from a government-appointed committee.

Minors & Consent Requirements

Minors under age 16 need parental consent. For minors, Data Fiduciary shall use technical and organizational measures for verification of the parent/lawful guardians' identity against reliable identity and age details.

Comparison with GDPR

The DPDP Act aligns with global trends in data privacy, mirroring core principles of the EU's General Data Protection Regulation (GDPR) such as explicit consent, data minimization, purpose limitation, rights to access, correction, and erasure. However, the DPDP Act's list of legitimate uses without consent is narrower than the GDPR's six lawful bases, which include a broad legitimate-interests basis; organisations that rely on GDPR legitimate interests to process data about Indian residents must map those activities to consent or a listed legitimate use.

Personal Data Processing Framework

The Act covers digital personal data, any data about an individual who is identifiable by or in relation to such data, in digital form, including data that was originally collected in non-digital form but has been digitised.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

28. assistant (2026-06-08T13:06:22.433Z)

I now have comprehensive data on all three sub-dimensions. The Gemma 4 Apache-2.0 claim is editorial assertion (not from a primary Google source) and noted it's image-multimodal only (no confirmed video), which is a relevant caveat. Let me do two final targeted checks: GPU-hour cost specifics for fine-tuning a 7B VLM (LoRA hours), and Modal/Predibase positioning plus the RTX 2070 local-training reality.

29. user (2026-06-08T13:06:29.142Z)

Web search results for query: "ActionFormer temporal action localization training cost GPU single small head TAL lightweight"

Links: [{"title":"TallFormer: Temporal Action Localization with a Long-Memory Transformer","url":"https://www.researchgate.net/publication/364634303_TallFormer_Temporal_Action_Localization_with_a_Long-Memory_Transformer"},{"title":"GitHub - wangcheng666/ActionFormer: Temporal Action Localization Enhanced through Cross-Modal and Cross-Structural Distillation · GitHub","url":"https://github.com/wangcheng666/ActionFormer"},{"title":"[2204.01680] TALLFormer: Temporal Action Localization with a Long-memory Transformer","url":"https://arxiv.org/abs/2204.01680"},{"title":"TallFormer: Temporal Action Localization with a Long-Memory Transformer | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-19830-4_29"},{"title":"TallFormer: Temporal Action Localization with a Long-memory Transformer","url":"https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136940495.pdf"},{"title":"TALLFormer: Temporal Action Localization with Long-memory Transformer | DeepAI","url":"https://deepai.org/publication/tallformer-temporal-action-localization-with-long-memory-transformer"},{"title":"TALLFormer: Temporal Action Localization with a Long-memory Transformer","url":"https://arxiv.org/pdf/2204.01680"}]

Based on the search results, here's what I found regarding ActionFormer temporal action localization and training cost:

Training Cost and GPU Requirements

Training ActionFormer on THUMOS requires ~4.5GB GPU memory, yet inference might require over 10GB GPU memory. A GPU with at least 12 GB of memory is recommended.

GPU Memory Challenges in Temporal Action Localization

The search results highlight a broader challenge in the field. Due to the high GPU memory cost caused by processing long untrimmed videos, many methods sacrifice the representational power of the short-term feature extractor by either freezing the backbone or using a small spatial video resolution. This issue becomes even worse with recent video transformer models, many of which have quadratic memory complexity.

Related Approaches to Address Training Costs

Research has addressed these issues through alternative architectures. For example, TALLFormer is a memory-efficient and end-to-end trainable Temporal Action Localization Transformer with Long-term memory that eliminates the need for processing hundreds of redundant video frames during each training iteration, thus significantly reducing GPU memory consumption and training time.

The results also mention modifications to transformer architectures, such as increasing the number of parallel attention heads from 8 to 16 in Multi-Head Self-Attention blocks when building on ActionFormer baseline models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

30. user (2026-06-08T13:06:30.839Z)

Web search results for query: "LoRA fine-tuning 7B vision language model GPU hours A100 H100 how long one epoch QLoRA VRAM"

Links: [{"title":"Fine-Tuning LLMs with LoRA: 2025 Guide | Amir Teymoori","url":"https://amirteymoori.com/fine-tuning-llms-with-lora-a-practical-guide-for-2025/"},{"title":"How to Fine-Tune LLMs with LoRA in Python ? A Complete Guide - machinelearningplus","url":"https://machinelearningplus.com/deep-learning/fine-tune-llms-with-lora-python/"},{"title":"How can I fine-tune large language models on a budget using LoRA and QLoRA on cloud GPUs?","url":"https://www.runpod.io/articles/guides/how-to-fine-tune-large-language-models-on-a-budget"},{"title":"Fine-Tune Mistral-7B with LoRA A Quickstart Guide | DigitalOcean","url":"https://www.digitalocean.com/community/tutorials/mistral-7b-fine-tuning"},{"title":"radio-llava: Advancing Vision-Language Models for Radio Astronomical Source Analysis","url":"https://arxiv.org/pdf/2503.23859"},{"title":"SARChat-Bench-2M: A Multi-Task Vision-Language Benchmark for SAR Image Interpretation","url":"https://arxiv.org/pdf/2502.08168"},{"title":"MMSci: A Dataset for Graduate-Level Multi-Discipline Multimodal   Scientific Understanding","url":"https://arxiv.org/pdf/2407.04903"},{"title":"Fine-Tuning on Noisy Instructions: Effects on Generalization and Performance","url":"https://arxiv.org/pdf/2510.03528"},{"title":"h2oGPT: Democratizing Large Language Models","url":"https://arxiv.org/pdf/2306.08161"},{"title":"Safety Fine-Tuning at (Almost) No Cost: A Baseline for Vision Large   Language Models","url":"https://arxiv.org/pdf/2402.02207"}]

Based on the search results, here's what I found about LoRA fine-tuning for 7B models and vision-language models:

Training Time for 7B Models

For 7B models with 1000 examples, training takes about 30-90 minutes on a single H100.

GPU Memory Requirements

For LoRA (16-bit):
You need about 2 GB of GPU memory per 1B parameters for model weights plus overhead, so a 7B model (~14 GB) with LoRA might need ~20 GB total.

For QLoRA (4-bit):
A 7B model in 4-bit is ~3.5 GB plus overhead ~8-10 GB, and in practice, 7B QLoRA can run on a 16 GB GPU comfortably.

Vision-Language Model Training Examples

For vision-language models specifically:
- Full fine-tuning of a 7B vision-language model required 4 NVIDIA A100 64 GB GPUs and approximately 15 hours per epoch on the combined training dataset.
- Fine-tuning a 2B model on a dataset took approximately 8 hours to complete on 8 NVIDIA A100 GPUs with 40GB each.

General Recommendations

Smaller 7-13B models train comfortably on a single 24GB consumer GPU, and for LoRA fine-tuning, a 7B model requires 16GB VRAM with 16-bit training.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

31. assistant (2026-06-08T13:06:35.491Z)

I have everything needed. Key findings: ActionFormer-class TAL heads train on ~4.5 GB VRAM (fits the RTX 2070 Super 8GB locally ? effectively $0 marginal), while a 7B VLM LoRA needs ~16-24 GB

VRAM (does NOT fit local; cloud-only). Let me do one final confirmation on the Qwen2.5-VL-72B being custom-licensed (already confirmed above) and synthesize. I have enough to deliver the structured output.

Let me compile the final structured answer.

32. user (2026-06-08T13:08:02.388Z)

Structured output provided successfully

33. assistant (2026-06-08T13:08:14.050Z)

Structured output delivered. Summary of what I found across the three sub-dimensions:

COST ? The two training paths differ by an order of magnitude. A small TAL/segmentation head (ActionFormer-class, ~4.5 GB training VRAM) fits the existing RTX 2070 Super locally at ~$0 marginal cost. A 2-7B video VLM LoRA (~16-24 GB VRAM) does NOT fit the 2070 and is cloud-only: ~$5-40 per run on rented GPU (RunPod A100 80GB ~$1.19-1.39/hr) or ~$2-30 on token-priced managed backends (Together/Fireworks LoRA SFT ?16B ~$0.48-0.50/M tokens). Full multimodal fine-tuning is ~10-15x heavier (~$70-300/epoch on 4xA100).

LICENSING ? Per-checkpoint verification matters most: Qwen2.5-VL-7B = Apache-2.0 (clean), but 3B = Qwen **Research** License (non-commercial) and 72B = custom Qwen License (100M-MAU gate + attribution). Qwen3-VL 4B/8B/30B = Apache-2.0. InternVL2.5 = MIT but some variants bundle a research-licensed Qwen-3B component. Gemma 3 rejection is well-grounded; Gemma 4 Apache-2.0 is AMBER (only secondary-source confirmation, image-multimodal only, no confirmed video). Training on owned footage + human labels is clean.

PRIVACY ? Local inference is the strongest lever (no cross-border issue). Fireworks and Together both contractually don't train on customer data (ZDR, SOC2 Type II, BAAs); Vertex won't without permission (Mumbai region available). India DPDP is residency-permissive (empty negative list as of Feb 2026) but requires verifiable parental consent for under-16s ? reinforcing the minors-exclusion guardrail.

Key caveats: GPU prices are volatile snapshots; the Gemma 4 Apache-2.0 + video claim needs primary-source confirmation; vendor "no-train" claims are self-asserted (get the signed DPA/BAA); the "own-footage-is-clean" reading is strong but not formal legal advice.

Relevant project file: `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md` (C14 row is the licensing/training-data gate this research feeds; §5 hosting unit-economics).